

Summary Report



1. Best Classification Model and Justification

The project builds an early-warning classification system for identifying high-risk customers. The notebook starts with a Logistic Regression baseline and then compares stronger tree-based models such as Decision Tree and Random Forest. For business use, the best final model should be the one that minimizes false negatives, because missing a high-risk customer is more costly than raising an alert for a low-risk customer.

Based on the workflow in the notebook, the best-performing model is the tuned Random Forest Classifier. Although Logistic Regression is a strong baseline, Random Forest is preferred because:

- it is strong for structured tabular data,
- it handles nonlinear relationships well,
- it can generalize better than a single Decision Tree,
- and its performance can be improved through hyperparameter tuning.

2. Impact of Imbalance Handling Techniques

The dataset is highly imbalanced, so the notebook evaluates several resampling techniques:

- Under-Sampling
- Over-Sampling
- SMOTE
- ADASYN

These methods help the model learn the minority class better. In imbalanced classification, models often become biased toward the majority class, which reduces recall for the high-risk class. The resampling methods improve the balance between classes and help the classifier focus on the minority class, which is important for risk detection.

3. Comparison of Performance Metrics

The notebook evaluates models using:

- Confusion Matrix
- Accuracy
- Precision
- Recall
- F1-Score
- AUC-ROC

For this project, recall and F1-score for the high-risk class are especially important. Accuracy alone is not enough because a model can have high overall accuracy while still missing most risky customers. The notebook therefore compares the models using recall for the minority class and AUC-ROC along with the confusion matrix.

4. Business Interpretation of False Positives and False Negatives

False Positive

- The model flags a low-risk customer as high-risk.
- Business impact: extra review effort, customer inconvenience, and possible unnecessary intervention.

False Negative

- The model misses a truly high-risk customer.
- Business impact: greater risk of fraud, default, or loss.

Because this is a risk classification problem, false negatives are more serious for the business. Therefore, the final model should prioritize recall for the high-risk class, even if it causes some additional false positives.